

Dept. of CSE

RIT Kottayam

S3 CSE - 2025

Data Structure Lab Cycles with Deadlines

Cycle No.	Experiments	Deadline
Cycle 1 (Polynomial Addition Using Array, Sparse Matrix Operations)	1. Write a program to read two polynomials and store them in an array. Calculate the sum of the two polynomials and display the first polynomial, second polynomial and the resultant polynomial. 2. Write a program to enter two matrices in normal form. Write a function to convert two matrices to tuple form and display it. Also, find the transpose of the two matrices represented in tuple form and display it. Find the sum of the two matrices in tuple form and display the sum in tuple form.	18/7/2025
Cycle 2 (Applications of stack, Queue)	1. Using stack, convert an infix expression to a postfix expression and evaluate the postfix expression. 2. Write a program to convert an infix expression to a prefix expression using stack. 3. Implement a Queue using arrays with the operations: a. Insert elements into the queue b. Delete elements from the queue c. Display the contents of the queue after each operation 4. Implement a Circular Queue using arrays with the operations: a. Insert elements into the queue b. Delete elements from the queue c. Display the contents of the queue after each operation 5. Implement a Double Ended Queue using arrays with the operations: a. Insert elements into the queue (both front and rear) b. Delete elements from the queue (both front and rear) c. Display the contents of the queue after each operation	18/8/2025
Cycle 3 (Linked Lists and Applications)	1. Implement a Queue using linked lists with the operations: a. Insert elements into the queue b. Delete elements from the queue c. Display the contents of the queue after each operation	27/8/2025
	2. Implement addition and multiplication of polynomials using singly linked lists.	
	3. Implement backward and forward navigation of visited web pages in a web browser (i.e. back and forward buttons) using doubly linked list operations.	
	4. Write a C program to simulate a basic memory allocator and garbage collector using doubly linked list	
Cycle 4 (Trees and Graphs)	1. Create a binary tree for a given simple arithmetic expression and find the prefix / postfix equivalent.	26/09/2025
	2. Implement a dictionary of word-meaning pairs using binary search trees.	
	3. Implement a Priority Queue using arrays with the operations: a. Insert elements into the queue b. Delete elements from the queue c. Display the contents of the queue after each operation.	
	4. Represent any given graph and perform depth first search and breadth first search.	